

Python for Data Science: *Vectorized* Thinking Workbook

Five hands-on drills that turn NumPy from syntax you copy into a mental model you own — loops to vectors, filters, and stats.

NAME

DATE

SPEEDUP ×

1 Time Your Own Loop-to-Vector Conversion

- Write a Python loop that **squares 50,000 integers** one at a time. Time it with `time`.
- Rewrite as **one vectorized line**: `arr ** 2`. Time that too.
- Divide loop time by vector time — your *speedup multiplier*. Tutorial saw 50–200x. Where do you land?

```
import time, numpy as np
arr = np.arange(50_000)
```

YOUR TIMINGS

2 Build the IQR Outlier Detector on Your Own Numbers

- Pull **10 real numbers** from your life — expenses, screen time, workout reps.
- Load into a NumPy array, run the tutorial's *six-line IQR block* exactly.
- Report which value(s) got **flagged** — does it match your gut sense of "weird day"?

```
q25, q75 = np.percentile(data, [25, 75])
iqr = q75 - q25
```

FLAGGED VALUES

3 Boolean Indexing Chain: From scores > 90 to Your Filter

- Build an array of **15 values** you can filter meaningfully — steps, scores, response times.
- Write **three chained filters** using `&` or `|`, matching `arr[(arr > x) | (arr < y)]`.
- Print each filtered result. Which condition returned the *fewest* matches?

```
arr[(arr > 90) | (arr < 60)]
arr[(arr >= 70) & (arr <= 85)]
```

FEWEST MATCHES

4 arange vs linspace: Prove They're Different Tools

- Generate **0 to 1** two ways: `np.arange(0, 1.1, 0.1)` and `np.linspace(0, 1, 11)`.
- Print both arrays side by side — look closely at the *endpoint* and array length.
- In one sentence: why does `arange`'s float drift make `linspace` the safer pick for an exact point count?

ONE-LINE ANSWER

5 dtype Detective: Break a Mixed-Type Array on Purpose

- Create `np.array([85, 92, 78])` — check `.dtype`.
- Add one float: `np.array([85, 92, 78.5])` — check `.dtype` again.
- Add one string: `np.array([85, '92', 78])` — check `.dtype` once more.
- Write down what happened to the **integers** each time, and why *one-type-per-array* forced it.

DTYPE RESULTS